DETAILED LITERATURE SURVEY FOR AIRSKETCH PROJECT

1. OMG-Bench: A New Challenging Benchmark for Skeleton-Based Online Micro Hand Gesture Recognition

Introduction

The paper "OMG-Bench: A New Challenging Benchmark for Skeleton-Based Online Micro Hand Gesture Recognition" addresses the growing need for efficient micro-gesture recognition systems in Human-Computer Interaction, Virtual Reality, and Augmented Reality applications. Existing gesture recognition datasets mainly focus on large hand movements and isolated gestures, which often lead to user fatigue and do not accurately represent real-world interaction scenarios. To overcome these limitations, the authors introduced OMG-Bench, the first large-scale benchmark specifically designed for skeleton-based online micro hand gesture recognition.

Methodology

The authors developed a multi-view RGB-D data acquisition system consisting of five synchronized cameras to capture hand movements from different angles. Using a self-supervised multi-view hand pose estimation pipeline, accurate three-dimensional hand skeletons were generated automatically. To reduce manual annotation effort, a semi-automatic labeling framework was proposed in which preliminary annotations were generated automatically and later refined by human experts. This approach enabled the creation of a large and accurately labeled dataset containing 40 micro-gesture classes and 13,948 gesture instances.

To address the challenges of recognizing subtle finger movements, the researchers proposed a novel recognition architecture called Hierarchical Memory-Augmented Transformer (HMATr). The model performs both gesture detection and classification simultaneously. HMATr utilizes hierarchical memory banks that store both short-term and long-term motion information, allowing the system to capture fine-grained finger movements and temporal dependencies within continuous gesture streams. Additionally, position-aware queries were introduced to improve temporal localization and distinguish highly similar gesture patterns.

Results and Findings

Experimental evaluation demonstrated that HMATr significantly outperformed existing state-of-the-art online gesture recognition methods. The benchmark highlighted key challenges in micro gesture recognition, including subtle inter-class differences, rapid motion patterns, and varying gesture durations. By combining accurate skeleton acquisition with memory-enhanced Transformer architectures, the proposed framework achieved improved detection rates and recognition accuracy while maintaining robustness in continuous gesture scenarios.

Advantages and Limitations

A major strength of this work is the introduction of the first large-scale benchmark for skeleton-based online micro gesture recognition along with a powerful Transformer-based recognition architecture. The multi-view acquisition system greatly improves skeleton quality and annotation accuracy. However, the framework requires specialized hardware and considerable computational resources, making deployment more challenging on low-power devices. Furthermore, performance may vary when applied to environments that differ significantly from the training conditions.

Relevance to AirSketch

This paper provided valuable insights for the AirSketch project, particularly regarding the importance of accurate hand landmark extraction and temporal gesture analysis. The study demonstrated how subtle finger movements can be captured and interpreted effectively using skeleton representations. The concepts of motion tracking, temporal context preservation, and robust gesture classification influenced the design of AirSketch, where precise hand tracking and continuous gesture interpretation are essential for achieving smooth and responsive virtual drawing functionality.

Conclusion

The OMG-Bench paper represents a significant contribution to gesture recognition research by introducing both a large-scale benchmark dataset and an advanced Transformer-based recognition framework. Its focus on accurate hand skeleton extraction, temporal modeling, and online recognition provides a strong foundation for modern gesture-controlled systems and offers several concepts that are directly applicable to the AirSketch project.

2. Improving the Performance of Unimodal Dynamic Hand Gesture Recognition with Multimodal Training

Introduction

Dynamic hand gesture recognition plays a vital role in Human-Computer Interaction, Virtual Reality, robotics, and intelligent control systems. Most modern gesture recognition systems rely on multiple data sources such as RGB images, depth maps, optical flow, and skeleton information to improve recognition accuracy. Although multimodal systems generally achieve better performance, they require additional sensors during deployment, increasing hardware complexity and cost. This paper proposes a novel framework that leverages multiple modalities during training while requiring only a single modality during testing, thereby combining the advantages of multimodal learning with the simplicity of unimodal deployment.

Methodology

The authors introduced a framework called Multimodal Training Unimodal Testing (MTUT), which allows multiple modalities to collaborate during the training phase. Separate neural networks are trained for each modality, including RGB, depth, and optical flow data. During training, these networks exchange information and learn from one another, enabling the model to capture richer gesture representations. However, during deployment, only a single modality is required, making the system practical for real-world applications.

To learn both spatial and temporal gesture information, the framework employs 3D Convolutional Neural Networks (3D-CNNs). Unlike traditional CNNs that analyze only spatial features, 3D-CNNs process both spatial and temporal dimensions simultaneously, allowing the model to understand gesture evolution across multiple frames. The paper also introduces a Spatiotemporal Semantic Alignment (SSA) mechanism that aligns feature representations learned from different modalities, ensuring effective knowledge transfer. To prevent excessive alignment and preserve modality-specific information, a Focal Regularization Parameter (FRP) is incorporated into the training process.

Results and Findings

The proposed MTUT framework was evaluated on several dynamic gesture recognition datasets containing RGB, depth, and motion information. Experimental results demonstrated that the framework consistently improved unimodal recognition accuracy compared to conventional single-modality approaches. The SSA mechanism successfully enhanced feature learning by transferring useful knowledge between modalities, while the FRP stabilized the training process and improved model convergence. As a result, the RGB-only model achieved performance closer to that of fully multimodal systems.

Advantages and Limitations

A major advantage of this work is its ability to improve recognition accuracy without requiring multiple sensors during deployment. The framework effectively utilizes multimodal information during training while maintaining a lightweight and cost-effective inference process. The use of 3D-CNNs also enables robust modeling of gesture dynamics. However, the training stage remains computationally expensive because it requires multiple synchronized modalities. In addition, collecting multimodal datasets can be time-consuming and may limit scalability in certain applications.

Relevance to AirSketch

This paper provided valuable insights into the importance of temporal information in gesture recognition systems. The use of 3D-CNNs and spatiotemporal feature extraction highlighted the need to analyze hand movement patterns rather than relying solely on static hand poses. This concept is particularly relevant to AirSketch, where drawing, color selection, and canvas control depend on continuous hand movements. The study also demonstrated how robust feature learning can improve performance even when only a single input source is available, aligning with AirSketch's webcam-based interaction approach.

Conclusion

The paper presents an innovative Multimodal Training Unimodal Testing framework that improves dynamic hand gesture recognition by exploiting multiple modalities during training while maintaining the simplicity of unimodal deployment. Through the use of 3D-CNNs, Spatiotemporal Semantic Alignment, and Focal Regularization, the proposed system achieves significant performance improvements and demonstrates an effective balance between recognition accuracy and deployment practicality. The concepts introduced in this work provide valuable guidance for developing efficient gesture-controlled applications such as AirSketch.

3. Dynamic LIBRAS Gesture Recognition via CNN over Spatiotemporal Matrix Representation

Introduction

Sign language recognition has become an important research area because it enables communication between hearing-impaired individuals and the wider community. Traditional gesture recognition systems often rely on specialized hardware such as data gloves, motion sensors, or depth cameras, which can be expensive and difficult to deploy. This paper proposes a dynamic LIBRAS (Brazilian Sign Language) gesture recognition system that uses only a standard camera, MediaPipe hand tracking, and a Convolutional Neural Network (CNN). The primary objective is to recognize dynamic hand gestures accurately while maintaining real-time performance and low computational requirements.

Methodology

The proposed system uses MediaPipe to detect and track 21 hand landmarks in each video frame. Instead of processing raw images directly, the authors extract the landmark coordinates and organize them into a Spatiotemporal Matrix Representation (SMR). This representation captures both the spatial structure of the hand and its movement over time, allowing gesture sequences to be analyzed more effectively.

To classify gestures, the researchers employ a Convolutional Neural Network that processes the spatiotemporal matrices and automatically learns meaningful gesture patterns. A sliding window mechanism is used to continuously analyze incoming frames, enabling real-time recognition without waiting for an entire gesture sequence to complete. The system also incorporates data augmentation techniques such as temporal frame duplication and sequence variation to improve model robustness and handle differences in gesture execution speed among users.

Results and Findings

The proposed framework was evaluated using a LIBRAS gesture dataset containing dynamic hand gestures performed by different users. Experimental results demonstrated that the combination of MediaPipe landmarks and CNN-based classification achieved high recognition accuracy while maintaining real-time performance. The spatiotemporal matrix representation effectively captured gesture dynamics, allowing the model to distinguish between gestures that appeared visually similar in individual frames but differed in movement patterns.

Advantages and Limitations

One of the major strengths of this approach is its ability to perform dynamic gesture recognition using only a standard RGB camera, eliminating the need for specialized hardware. The use of landmark coordinates significantly reduces computational complexity compared to image-based approaches, while the CNN efficiently learns gesture patterns automatically. However, system performance remains dependent on the accuracy of MediaPipe hand tracking. Severe hand occlusion, poor lighting conditions, and highly complex gestures may reduce recognition accuracy and require further improvements.

Relevance to AirSketch

This paper is highly relevant to the AirSketch project because both systems rely heavily on MediaPipe hand tracking and real-time gesture interpretation. The use of 21 hand landmarks demonstrated how accurate hand representations can be obtained without processing full image frames. The concept of analyzing continuous motion rather than individual frames influenced the gesture interpretation techniques used in AirSketch, where drawing operations depend on fingertip movement over time. Additionally, the study reinforced the importance of efficient landmark-based representations for achieving responsive and computationally efficient interaction systems.

Conclusion

The paper presents an effective approach for dynamic sign language recognition by combining MediaPipe hand tracking, spatiotemporal matrix representation, and CNN-based classification. The proposed system successfully captures both spatial and temporal gesture information while maintaining real-time performance. Its emphasis on landmark-based feature extraction and continuous gesture analysis provides valuable insights for the development of gesture-controlled applications such as AirSketch, where accurate hand tracking and smooth user interaction are essential.

4. Vision Based Hand Gesture Recognition for Human Computer Interaction: A Survey

Introduction

Hand gesture recognition has emerged as one of the most promising technologies for Human-Computer Interaction (HCI) because it enables users to communicate with computers in a natural and intuitive manner. Traditional input devices such as keyboards and mice create a physical barrier between humans and machines, whereas gesture-based systems allow interaction through natural hand movements. This survey paper provides a comprehensive review of vision-based hand gesture recognition techniques, their underlying technologies, applications, challenges, and future research directions. The authors analyze various approaches developed over the years and highlight the growing importance of gesture recognition in modern interactive systems.

Methodology

The survey categorizes gesture recognition systems into contact-based and vision-based approaches. Contact-based systems use specialized hardware such as data gloves, accelerometers, and motion sensors to capture hand movements with high accuracy. However, these systems are often expensive and uncomfortable for users. In contrast, vision-based systems utilize cameras and image processing algorithms to detect, track, and recognize hand gestures without requiring physical devices, making them more practical for real-world applications.

The authors further describe the gesture recognition process as consisting of three major stages: hand detection, hand tracking, and gesture recognition. Various techniques are reviewed for each stage, including skin color segmentation, background subtraction, contour extraction, optical flow analysis, Kalman filters, particle filters, neural networks, Hidden Markov Models (HMMs), Support Vector Machines (SVMs), and fuzzy logic-based classifiers. The survey also discusses different gesture representations, including appearance-based and model-based approaches, and examines the strengths and limitations of each technique.

Results and Findings

The survey concludes that vision-based gesture recognition systems provide a more natural and user-friendly interaction method compared to traditional interfaces. The authors observe that advancements in computer vision and machine learning have significantly improved recognition accuracy and enabled real-time performance. The study also highlights the increasing adoption of gesture recognition in applications such as virtual reality, robotics, gaming, healthcare, sign language interpretation, surveillance, and intelligent control systems.

Advantages and Limitations

A major advantage of vision-based gesture recognition is that it enables touchless interaction without requiring expensive hardware. These systems are generally more accessible, flexible, and scalable than contact-based alternatives. However, the survey identifies several challenges including sensitivity to lighting variations, background complexity, hand occlusion, computational requirements, and difficulties in recognizing large gesture vocabularies. These limitations continue to motivate ongoing research in the field.

Relevance to AirSketch

This survey paper provided the theoretical foundation for the AirSketch project by explaining the complete workflow of gesture recognition systems, including detection, tracking, and recognition stages. The discussion of vision-based approaches validated the decision to use a webcam-based interaction system instead of specialized hardware. Additionally, the challenges identified in the survey, such as illumination changes and hand occlusion, closely resemble the practical issues encountered during AirSketch development. The paper also highlighted the growing role of gesture-based interfaces in creating intuitive and accessible Human-Computer Interaction systems, which aligns directly with the objectives of AirSketch.

Conclusion

The survey presents a detailed overview of vision-based hand gesture recognition technologies and their applications in Human-Computer Interaction. By reviewing a wide range of detection, tracking, and classification techniques, the paper establishes a strong theoretical understanding of gesture recognition systems. The concepts discussed provide valuable insights into the design and implementation of modern gesture-controlled applications and serve as an important foundation for the development of AirSketch.

5. Painting with Hand Gestures using MediaPipe

Introduction

The increasing popularity of touchless interaction systems has encouraged researchers to explore gesture-based alternatives to traditional input devices such as keyboards, mice, and touchscreens. The paper "Painting with Hand Gestures using MediaPipe" presents a virtual drawing application that allows users to create digital artwork through hand movements captured by a webcam. By combining computer vision and machine learning technologies, the system enables users to draw naturally in the air without requiring physical contact with any device. The study demonstrates the practical application of hand gesture recognition in creating intuitive and interactive Human-Computer Interaction systems.

Methodology

The proposed system utilizes OpenCV and MediaPipe as its primary technologies. OpenCV is responsible for image processing, video capture, frame manipulation, and rendering operations, while MediaPipe performs real-time hand detection and tracking. The framework identifies 21 hand landmarks and continuously tracks finger positions throughout the video stream. Using these landmarks, the system detects the index fingertip and records its movement trajectory to create drawing strokes on a virtual canvas.

The workflow begins with webcam frame acquisition, followed by hand detection and landmark extraction. The system then analyzes finger positions to identify specific gestures corresponding to drawing operations, color selection, brush control, and canvas clearing. By continuously updating the fingertip coordinates, the application creates a smooth drawing experience that closely resembles writing or sketching on a physical surface. The use of machine learning-based landmark detection significantly improves tracking accuracy compared to traditional contour-based methods.

Results and Findings

The experimental results demonstrated that the system could accurately detect and track hand movements in real time while maintaining smooth drawing performance. MediaPipe provided reliable hand landmark detection under normal operating conditions, enabling effective gesture recognition and responsive canvas interaction. The study showed that combining OpenCV with MediaPipe creates an efficient and practical framework for developing touchless drawing applications.

Advantages and Limitations

One of the major strengths of the proposed system is its ability to provide a completely touchless drawing experience using only a standard webcam. The approach is cost-effective, easy to deploy, and requires minimal hardware resources. The integration of machine learning improves tracking robustness and reduces dependence on handcrafted image processing techniques. However, the system remains sensitive to lighting conditions, camera quality, and hand occlusion. Rapid hand movements may also introduce tracking errors, affecting drawing accuracy in certain situations.

Relevance to AirSketch

This paper is highly relevant to the AirSketch project because both systems share similar objectives and technological foundations. The use of MediaPipe for hand landmark detection directly influenced the hand-tracking mechanism adopted in AirSketch. The concepts of fingertip-based drawing, gesture-controlled tool selection, and virtual canvas interaction provided valuable guidance during system development. Additionally, the paper demonstrated how real-time gesture recognition can be effectively integrated into a drawing application, reinforcing many of the design decisions implemented in AirSketch.

Conclusion

The paper presents an effective gesture-controlled virtual painting system that combines OpenCV and MediaPipe to enable touchless drawing through hand movements. By utilizing real-time hand tracking and machine learning-based landmark detection, the system successfully creates an intuitive and interactive drawing environment. The methodologies and architectural concepts discussed in this work served as an important reference for the development of AirSketch and highlighted the potential of gesture-based interfaces in modern Human-Computer Interaction applications.

6. Air Canvas Hand Tracking

Introduction

The increasing demand for touchless Human-Computer Interaction systems has encouraged the development of gesture-based applications that replace traditional input devices. The paper "Air Canvas Hand Tracking" presents a virtual drawing system that allows users to create drawings in free space using hand gestures captured through a webcam. By combining Artificial Intelligence, Computer Vision, OpenCV, and MediaPipe, the proposed system enables users to interact with a digital canvas without requiring physical contact with a mouse, stylus, or touchscreen. The study demonstrates how hand tracking technologies can be applied to create intuitive and accessible drawing applications.

Methodology

The proposed system is implemented using Python, OpenCV, and MediaPipe. OpenCV is responsible for capturing video frames, image processing, and rendering the virtual canvas, while MediaPipe provides real-time hand detection and tracking capabilities. The framework detects 21 hand landmarks and continuously tracks fingertip movements to determine user actions.

The system follows a structured workflow beginning with video acquisition through a webcam. The captured frames are processed using OpenCV and passed to MediaPipe for hand landmark extraction. The extracted landmark coordinates are analyzed to identify gestures and fingertip positions, which are then translated into drawing operations. By connecting consecutive fingertip coordinates, the system generates continuous drawing strokes on a virtual canvas. The architecture supports real-time interaction, allowing users to draw naturally in the air while receiving immediate visual feedback.

Results and Findings

The experimental results demonstrated that the system could successfully track hand movements and convert them into virtual drawings with minimal latency. MediaPipe provided accurate landmark detection and stable hand tracking, enabling smooth interaction between the user and the digital canvas. The study confirmed that computer vision-based hand tracking can effectively replace conventional drawing devices in many scenarios while maintaining an intuitive user experience.

Advantages and Limitations

One of the key advantages of the proposed system is its ability to provide touchless interaction using only a standard webcam and software-based processing. The framework is cost-effective, easy to deploy, and offers a natural drawing experience without requiring specialized hardware. However, the system remains sensitive to environmental factors such as lighting conditions, camera positioning, background complexity, and hand occlusion. Tracking accuracy may also decrease when gestures are performed too quickly or when the hand moves outside the camera's field of view.

Relevance to AirSketch

Among all the surveyed papers, this work is one of the most closely related to the AirSketch project because both systems focus on virtual drawing through hand gesture recognition. The use of MediaPipe as the primary hand-tracking framework directly influenced the implementation of AirSketch. The concepts of fingertip tracking, gesture interpretation, and real-time canvas interaction provided valuable guidance during development. Additionally, the challenges identified in the paper, including lighting variations and tracking limitations, closely resemble the practical issues encountered while designing AirSketch.

Conclusion

The Air Canvas Hand Tracking paper presents an effective virtual drawing system that combines OpenCV and MediaPipe to enable gesture-controlled interaction. Through accurate hand tracking and real-time processing, the system allows users to create drawings naturally without relying on traditional input devices. The methodologies and architectural concepts introduced in this work significantly contributed to the design and implementation of AirSketch, particularly in the areas of hand tracking, gesture recognition, and virtual canvas management.

7. Hand Gesture Recognition using Image Processing and Feature Extraction Techniques

Introduction

Hand gesture recognition has become an important area of research in Human-Computer Interaction, Computer Vision, and Artificial Intelligence. Gesture-based systems allow users to communicate with computers naturally without relying on traditional input devices such as keyboards and mice. This paper focuses on recognizing hand gestures using image processing and feature extraction techniques, emphasizing the importance of identifying meaningful hand characteristics before classification. The study demonstrates how carefully extracted features can improve recognition accuracy while reducing computational complexity.

Methodology

The proposed system follows a traditional image-processing-based approach consisting of image acquisition, preprocessing, segmentation, feature extraction, and classification. Initially, hand images are captured using a camera and undergo preprocessing operations such as noise removal, image enhancement, and color space conversion. The hand region is then separated from the background using segmentation techniques including skin color detection, thresholding, edge detection, and contour extraction.

Once the hand region is isolated, various geometric and shape-based features are extracted. These include hand contours, convex hulls, convexity defects, fingertip positions, hand orientation, and palm characteristics. The extracted features provide a compact numerical representation of hand gestures, which is then used for classification. The paper discusses the use of machine learning algorithms such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees to recognize different gesture categories.

Results and Findings

The study demonstrates that feature extraction significantly improves recognition efficiency by reducing the amount of information that must be processed. Experimental results showed that geometric and shape-based features are effective for distinguishing different hand gestures while maintaining relatively low computational requirements. The authors conclude that carefully selected features play a crucial role in improving classification accuracy and system performance.

Advantages and Limitations

One of the primary advantages of this approach is its simplicity and low computational cost compared to deep learning-based systems. Feature extraction techniques reduce processing requirements and make real-time gesture recognition feasible on devices with limited resources. However, the system remains sensitive to lighting variations, background complexity, skin tone differences, and hand occlusion. Additionally, handcrafted feature extraction methods often require extensive tuning and may struggle with complex gesture sets compared to modern machine learning frameworks.

Relevance to AirSketch

This paper provided important theoretical knowledge regarding traditional gesture recognition techniques and feature engineering. The concepts of fingertip detection, contour analysis, convex hull computation, and hand segmentation helped establish a foundation for understanding gesture recognition before adopting MediaPipe-based solutions. Although AirSketch primarily relies on machine learning-based hand landmark detection rather than handcrafted features, the paper highlighted the importance of identifying meaningful hand characteristics for accurate gesture interpretation. The study also reinforced the need for efficient processing to achieve smooth real-time interaction.

Conclusion

The paper presents a comprehensive approach to hand gesture recognition using image processing and feature extraction techniques. By combining preprocessing, segmentation, feature extraction, and machine learning classification, the system effectively recognizes hand gestures while maintaining computational efficiency. Although modern frameworks such as MediaPipe have largely automated feature extraction, the principles discussed in this work remain fundamental to understanding gesture recognition systems and provided valuable insights during the development of AirSketch.

8. Hand Gesture and Voice-Controlled Mouse for Physically Challenged Using Computer Vision

Introduction

Human-Computer Interaction has evolved significantly with the introduction of computer vision and artificial intelligence technologies. Traditional input devices such as keyboards and mice may not always be suitable for individuals with physical disabilities or users seeking more natural interaction methods. This paper presents a virtual mouse system that combines hand gesture recognition and voice commands to control computer operations without requiring physical hardware. The proposed system aims to improve accessibility, enhance user convenience, and provide an efficient touchless interaction platform.

Methodology

The proposed system utilizes computer vision, deep learning, and voice recognition technologies to interpret user actions. A webcam is used to capture hand gestures, while MediaPipe is employed for real-time hand detection and landmark extraction. The extracted hand features are processed using a deep learning model based on EfficientNet-B4, which classifies gestures and maps them to specific mouse operations such as cursor movement, clicking, scrolling, and drag-and-drop actions.

In addition to gesture recognition, the system integrates a voice assistant module that allows users to perform tasks through spoken commands. Voice commands are used for actions such as opening applications, searching information, retrieving system details, and controlling gesture recognition functionality. The combination of gesture and voice inputs creates a multimodal interaction system capable of providing a more flexible and accessible user experience.

Results and Findings

Experimental evaluation showed that the system successfully performed mouse control operations through hand gestures while maintaining reliable recognition accuracy. The integration of EfficientNet-B4 improved classification performance, and MediaPipe provided stable hand tracking in real-time environments. The voice assistant module further enhanced system usability by allowing users to perform additional computer operations without relying solely on gestures.

Advantages and Limitations

One of the main strengths of the proposed system is its ability to replace traditional mouse devices using only a webcam and microphone, making it particularly beneficial for physically challenged users. The multimodal design improves flexibility and accessibility while maintaining relatively low hardware requirements. However, system performance may be affected by lighting conditions, camera quality, background complexity, and gesture execution variations. Recognition delays and environmental noise may also influence the effectiveness of voice-based commands.

Relevance to AirSketch

This paper provided valuable insights into gesture-to-action mapping and real-time hand tracking using MediaPipe. Although the primary objective of the system is mouse control rather than virtual drawing, the underlying concepts of interpreting hand gestures and translating them into computer actions closely resemble the functionality of AirSketch. The study also highlighted the importance of designing intuitive gestures and maintaining responsive interaction, both of which are critical requirements for a gesture-controlled drawing application. Furthermore, the integration of voice commands suggests potential future enhancements that could be incorporated into AirSketch.

Conclusion

The paper presents a multimodal Human-Computer Interaction system that combines hand gesture recognition and voice commands to create a virtual mouse. By utilizing MediaPipe, EfficientNet-B4, and speech recognition technologies, the system successfully enables touchless computer control while improving accessibility and user convenience. The methodologies discussed in this work provide useful guidance for developing gesture-based interactive systems and contributed valuable ideas toward the design and implementation of AirSketch.

9. MediaPipe: A Framework for Building Perception Pipelines

Introduction

Modern computer vision applications such as hand tracking, face detection, pose estimation, and gesture recognition require efficient processing of large amounts of visual data in real time. Developing such systems from scratch is often challenging due to the complexity of machine learning integration, hardware optimization, and cross-platform deployment. To address these issues, Google Research introduced MediaPipe, an open-source framework designed for building real-time perception pipelines. The framework has become widely used in computer vision applications because of its ability to provide accurate landmark detection and efficient machine learning inference across multiple platforms.

Methodology

MediaPipe is built around a graph-based architecture in which data flows through interconnected processing nodes known as calculators. Each calculator performs a specific task such as image preprocessing, landmark detection, classification, or rendering. The framework uses packets and streams to manage data flow efficiently and allows multiple processing stages to operate in parallel. This modular architecture enables developers to build scalable perception systems while maintaining high performance.

One of the most important components of MediaPipe is the MediaPipe Hands solution, which performs real-time hand detection and tracking. The framework uses a two-stage pipeline consisting of palm detection followed by hand landmark estimation. Once a hand is detected, MediaPipe predicts 21 three-dimensional hand landmarks representing finger joints, fingertips, and palm positions. The tracking mechanism uses information from previous frames to reduce computational overhead and maintain smooth interaction, making it highly suitable for gesture recognition applications.

Results and Findings

The framework demonstrated excellent performance across a variety of computer vision tasks, including hand tracking, face mesh estimation, and pose detection. MediaPipe achieved accurate landmark estimation while maintaining real-time processing speeds on mobile devices, desktop systems, and web platforms. Its efficient tracking mechanism significantly reduced latency and improved user experience, making it one of the most widely adopted frameworks for gesture-based applications.

Advantages and Limitations

MediaPipe offers several advantages, including real-time performance, cross-platform compatibility, modular architecture, GPU acceleration, and seamless integration with machine learning models. These features simplify the development of perception-based applications and reduce implementation effort. However, the framework may experience reduced accuracy under poor lighting conditions, severe hand occlusion, or complex multi-hand interactions. Additionally, advanced customization requires a deeper understanding of its graph-based architecture.

Relevance to AirSketch

This paper is one of the most important references for the AirSketch project because MediaPipe serves as the core technology used for hand tracking and gesture recognition. The 21-hand-landmark model forms the foundation for identifying fingertip positions, drawing gestures, and user interactions within the virtual canvas. The framework's real-time tracking capabilities help maintain smooth and responsive drawing performance, while its browser compatibility supports the web-based implementation of AirSketch. The study demonstrated how accurate hand tracking can be achieved without requiring custom machine learning models, significantly simplifying system development.

Conclusion

MediaPipe provides a powerful and flexible framework for building real-time perception systems through its graph-based architecture and machine learning integration capabilities. Its ability to perform accurate hand tracking, landmark estimation, and efficient processing has made it a leading solution for gesture recognition applications. The concepts and technologies introduced in this framework played a crucial role in the design and implementation of AirSketch, particularly in the areas of hand detection, gesture interpretation, and real-time interaction.

10. TensorFlow.js: Machine Learning for the Web and Beyond

Introduction

The growing demand for intelligent web applications has created a need for machine learning frameworks that can operate directly within web browsers. Traditionally, machine learning models were executed on servers or native applications, requiring users to send data over the internet for processing. This approach often introduced latency, privacy concerns, and infrastructure costs. TensorFlow.js was developed to address these limitations by enabling machine learning development, deployment, and inference directly within JavaScript environments. The framework allows developers to build AI-powered web applications capable of performing real-time processing without relying on external servers.

Methodology

TensorFlow.js extends the TensorFlow ecosystem to JavaScript and provides a complete set of tools for developing and deploying machine learning models. The framework supports tensor-based computation, neural network development, model training, and inference within web browsers. It also includes a model conversion mechanism that allows models created in TensorFlow or Keras to be converted into formats suitable for browser-based deployment.

To achieve efficient performance, TensorFlow.js utilizes hardware acceleration through technologies such as WebGL and WebGPU. These technologies enable the framework to execute complex machine learning operations on graphics processing units, significantly improving computation speed. The framework also supports integration with computer vision systems and machine learning libraries, making it suitable for applications involving image classification, object detection, gesture recognition, and real-time video analysis.

Results and Findings

The framework demonstrated that machine learning models can be executed efficiently within web browsers while maintaining acceptable performance levels. Browser-based inference reduced latency by eliminating the need for server communication and improved privacy by processing data locally on the user's device. The use of GPU acceleration enabled real-time execution of machine learning tasks, making TensorFlow.js suitable for interactive applications that require immediate feedback.

Advantages and Limitations

One of the major strengths of TensorFlow.js is its ability to bring machine learning directly to the web while maintaining cross-platform compatibility. The framework reduces deployment complexity, improves user privacy, and supports offline functionality. Its integration with modern web technologies makes it highly accessible to developers. However, browser environments impose memory and performance limitations, and large machine learning models may experience slower execution compared to native implementations. Performance can also vary depending on the user's hardware and browser capabilities.

Relevance to AirSketch

TensorFlow.js is highly relevant to the AirSketch project because it enables machine learning functionality within a browser-based environment. The framework supports real-time processing and integrates effectively with MediaPipe, allowing gesture recognition systems to operate without requiring external servers. Its browser compatibility aligns with the web-based deployment strategy adopted in AirSketch and contributes to smooth, low-latency user interaction. The concepts of client-side inference and hardware-accelerated processing helped demonstrate how advanced AI features can be incorporated into interactive web applications.

Conclusion

TensorFlow.js represents a significant advancement in web-based machine learning by enabling model execution directly within JavaScript environments. Through support for neural networks, hardware acceleration, model conversion, and real-time inference, the framework has made AI accessible to modern web applications. Its ability to provide efficient client-side processing and seamless integration with computer vision systems makes it a valuable technology for gesture-controlled applications such as AirSketch, where responsiveness and accessibility are essential requirements.

Summary and Key Findings

The literature survey conducted for the AirSketch project provided a comprehensive understanding of the technologies, methodologies, and research developments in the fields of hand gesture recognition, computer vision, machine learning, and Human-Computer Interaction. The reviewed papers covered a broad spectrum of topics ranging from traditional image processing techniques to modern deep learning frameworks, offering valuable insights into the design and implementation of gesture-controlled systems.

The survey revealed that gesture recognition has evolved significantly over the years. Earlier approaches primarily relied on image processing techniques such as skin color segmentation, contour extraction, convex hull analysis, and handcrafted feature extraction methods. Although these techniques were computationally efficient, they were often sensitive to lighting conditions, background complexity, and hand occlusion. Recent advancements have shifted towards machine learning and deep learning-based solutions that provide improved accuracy, robustness, and real-time performance.

Several important findings emerged from the reviewed literature:

- Accurate hand tracking is the foundation of any gesture recognition system, and frameworks such as MediaPipe have significantly simplified this process through reliable hand landmark detection.
- Temporal information plays a crucial role in dynamic gesture recognition. Studies such as OMG-Bench and multimodal gesture recognition demonstrated that analyzing gesture movement over time produces better results than relying solely on static hand poses.
- Landmark-based approaches are more efficient than processing entire image frames because they reduce computational complexity while preserving important gesture information.
- Deep learning models such as CNNs and Transformer-based architectures improve recognition accuracy by automatically learning gesture patterns and temporal dependencies.

- Vision-based systems are generally preferred over wearable sensor-based systems because they provide a more natural, cost-effective, and user-friendly interaction experience.
- Real-time processing is a critical requirement for interactive applications such as virtual drawing systems, virtual mice, and gesture-controlled interfaces.
- Browser-based machine learning frameworks such as TensorFlow.js make it possible to deploy intelligent applications directly on the web without requiring dedicated servers or specialized hardware.

The survey also identified several common challenges faced by gesture recognition systems, including variations in lighting conditions, hand occlusion, background interference, gesture ambiguity, and differences in user behavior. These limitations continue to be active research areas and influence the design of modern gesture-controlled applications.

From the perspective of the AirSketch project, the reviewed papers provided valuable guidance for selecting technologies and designing system architecture. MediaPipe was identified as the most suitable framework for hand tracking due to its accuracy and real-time performance, while TensorFlow.js demonstrated the feasibility of browser-based deployment. The virtual painting and Air Canvas studies offered practical insights into gesture-controlled drawing systems, and the remaining papers provided a strong theoretical foundation for understanding gesture recognition methodologies and Human-Computer Interaction principles.

In conclusion, the literature survey established a clear understanding of current research trends and technological advancements in gesture recognition systems. The knowledge gained from these studies directly influenced the development of AirSketch and helped in designing a responsive, efficient, and user-friendly virtual drawing platform. The reviewed literature confirms that modern computer vision and machine learning technologies have made touchless interaction systems increasingly practical, accessible, and effective for real-world applications.